

The Novo Nordisk Foundation Biotechnology Research Institute for the Green Transition, DTU

Transcriptomics Training

Introduction to Illumina short-read RNA sequencing, bioinformatics and data analysis

**The Novo Nordisk Foundation Biotechnology Research Institute for the Green Transition
Technical University of Denmark**

Data Science Platform

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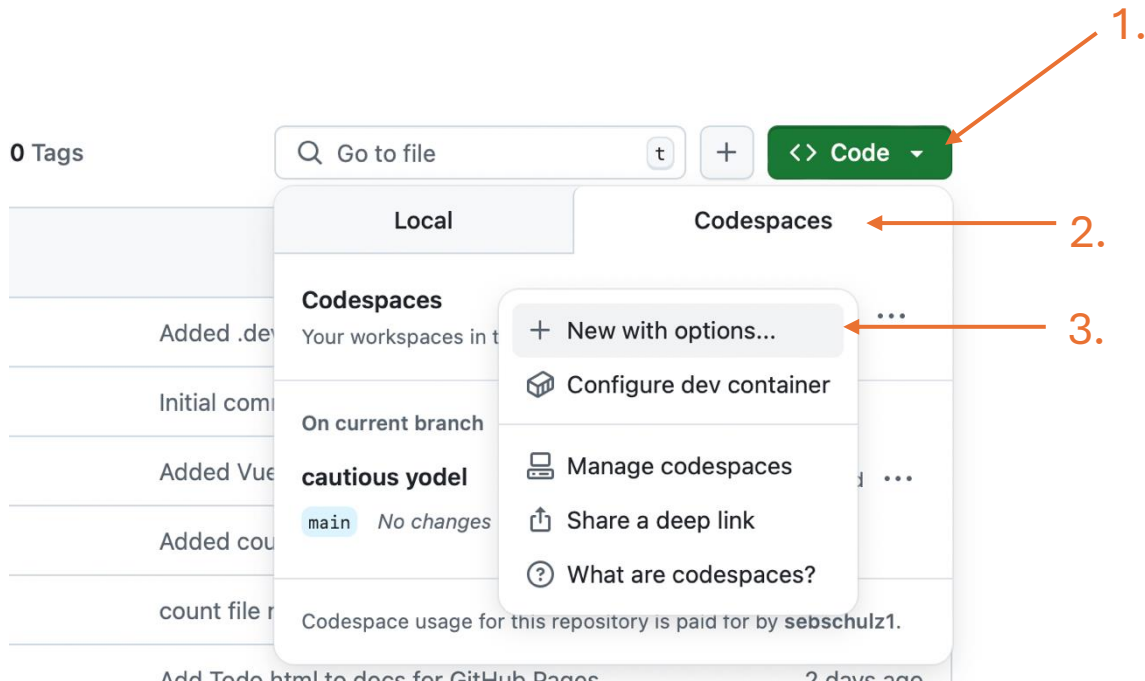
Albert Pallejà Caro (Head of Data Science Platform)

Agenda – 9.00-16.30

- **9.00:** Arrival and coffee
- **9.15: Session 1**
 - Short course intro (Albert) – **5 min**
 - Theoretical background on RNAseq, incl. relevant molecular biology, and data processing (Sebastian) – **75 min** (24 slides)
- *~10:30: Coffee break - 15 min*
- **10.45: Session 2**
 - Case study intro, hands-on data processing using the nf-core/rnaseq pipeline (Sebastian) – **25 min** (2 slides + code)
 - Evaluation of RNAseq processing results (multiQC report) – **25 min** (Sebastian)
 - Theoretical background on RNAseq data analysis (Sebastian) – **45 min** (12 slides)
- *~12.15: Lunch – 30 min*
- **13.00: Session 3**
 - Hands-on RNAseq data analysis using R (Juliana) – **75 min**
- *~14.15: Coffee break – 15 min*
- **14.30: Session 3 - continued**
 - Hands-on RNAseq data analysis using R (Juliana) – **60 min**
- *~15.30: Coffee break – 15 min*
- **15.45: Session 4**
 - Hands-on RNAseq data analysis using R (Juliana) – **30 min**
- *16.30: Wrap and Q&A Course ends*

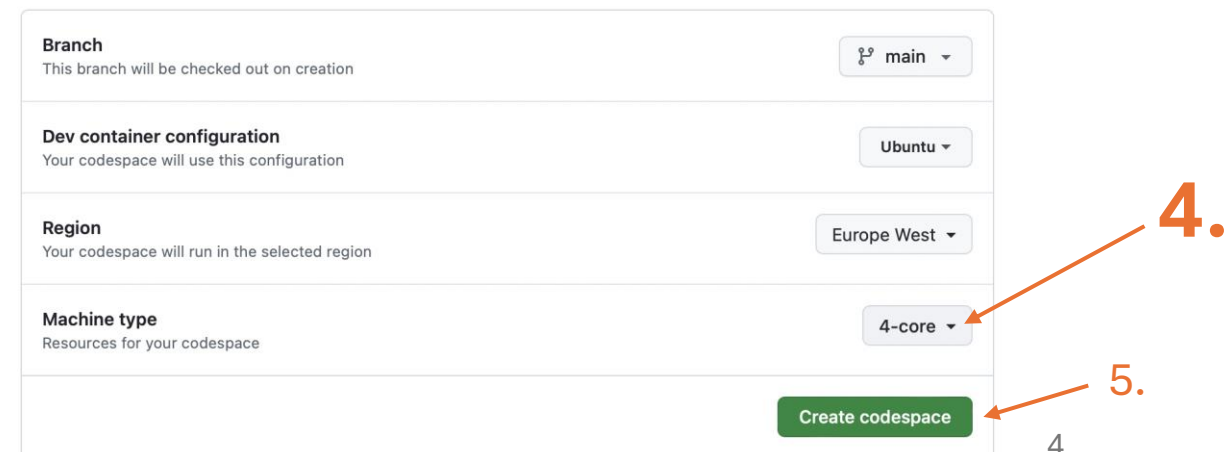
Create the codespace

- Log into your github account
- Open the training repo: https://github.com/biosustain/dsp_transcriptomics_training
- Create codespace **with 4 cores** which we will use to run code
- Creating the codspace will take about 30 mins



!!! IMPORTANT !!!
Select **4-cores** for machine type before creating the codespace

Create codespace for
biosustain/dsp_transcriptomics_training



Course overview and your questions

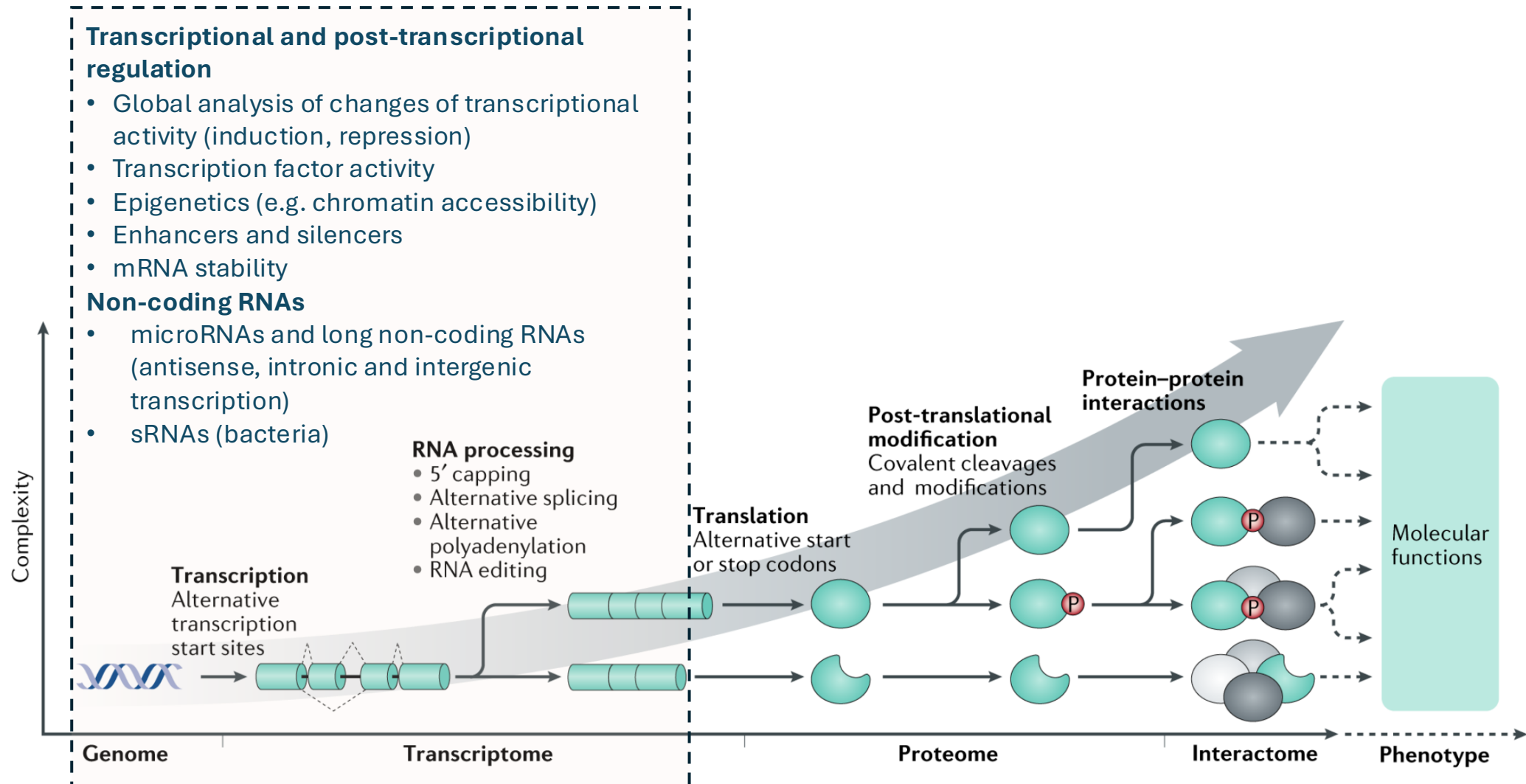
Topics of this introductory course

- Introduction to Illumina short read RNA sequencing
 - Theoretical background
 - Data processing
 - Standardized nf-core/rnaseq pipeline
 - Relevant software components
 - Evaluation of results (quality control)
 - Downstream data analysis
 - Differential gene expression analysis of a treatment-control study
 - Statistics
 - Basic methods for biological interpretation
 - Visualization

Your questions

- | | |
|--|--------------------------|
| • General overview of RNAseq processing and analysis | covered |
| • Required software tools | covered |
| • Visualization | covered |
| • iModulon analysis | not covered (discussion) |
| • Time-course RNAseq analysis | not covered (discussion) |
| • Single-cell RNAseq | not covered (discussion) |

Studying the transcriptome



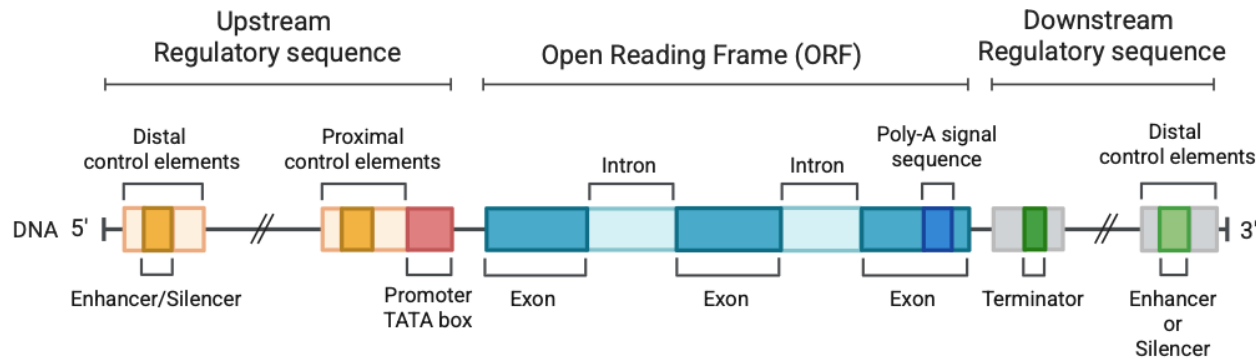
Eukaryotic example shown here

Modified from <https://doi.org/10.1038/s41580-020-0231-2>

Inspired by Marco Reverenna, Proteomics course 2025 ([Slides on github](#))

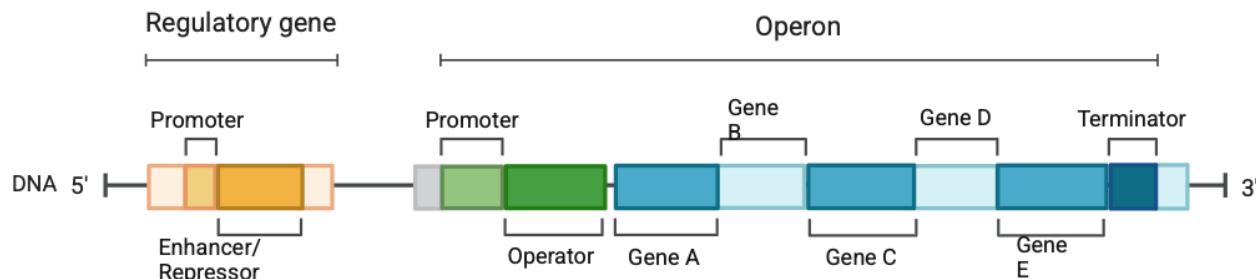
Eukaryotic and prokaryotic gene structure

Eukaryotic Gene Structure



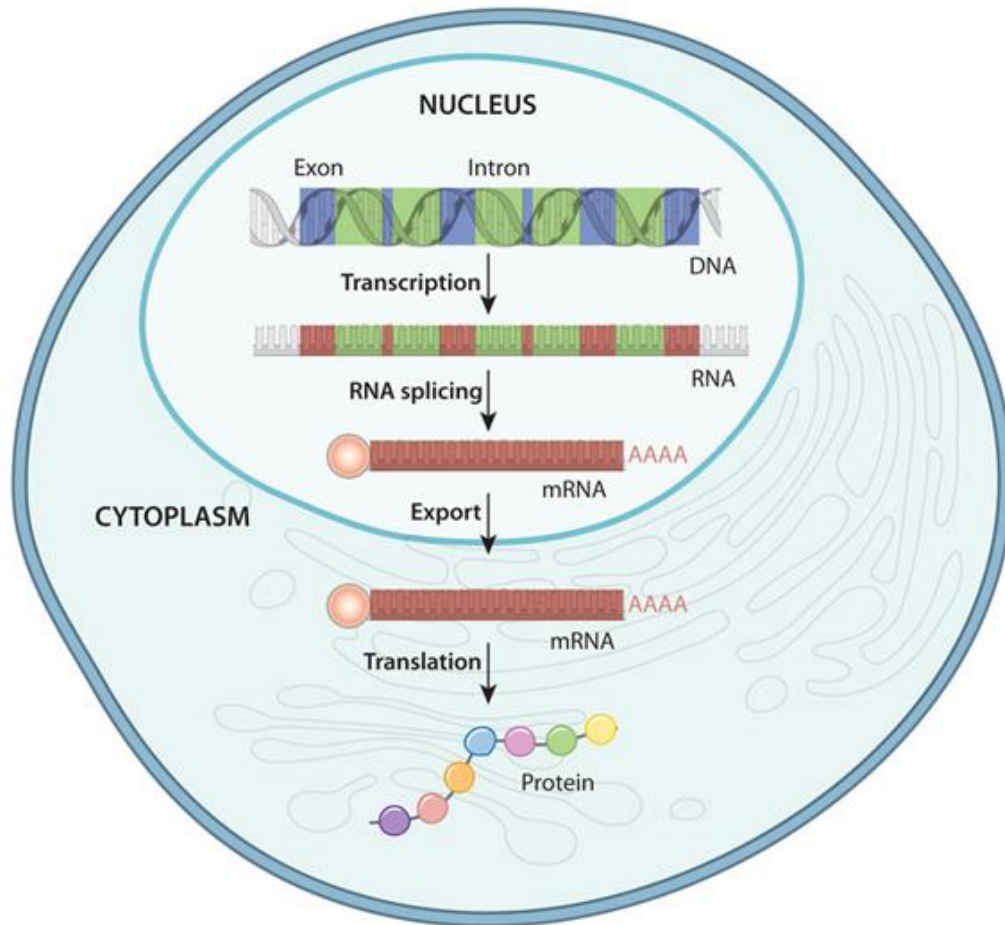
- Mostly **monocistronic**
- Exons and introns
- **Impact on data analysis:** align reads against exons only if the goal is to study gene expression of protein-coding genes

Prokaryotic Gene Structure



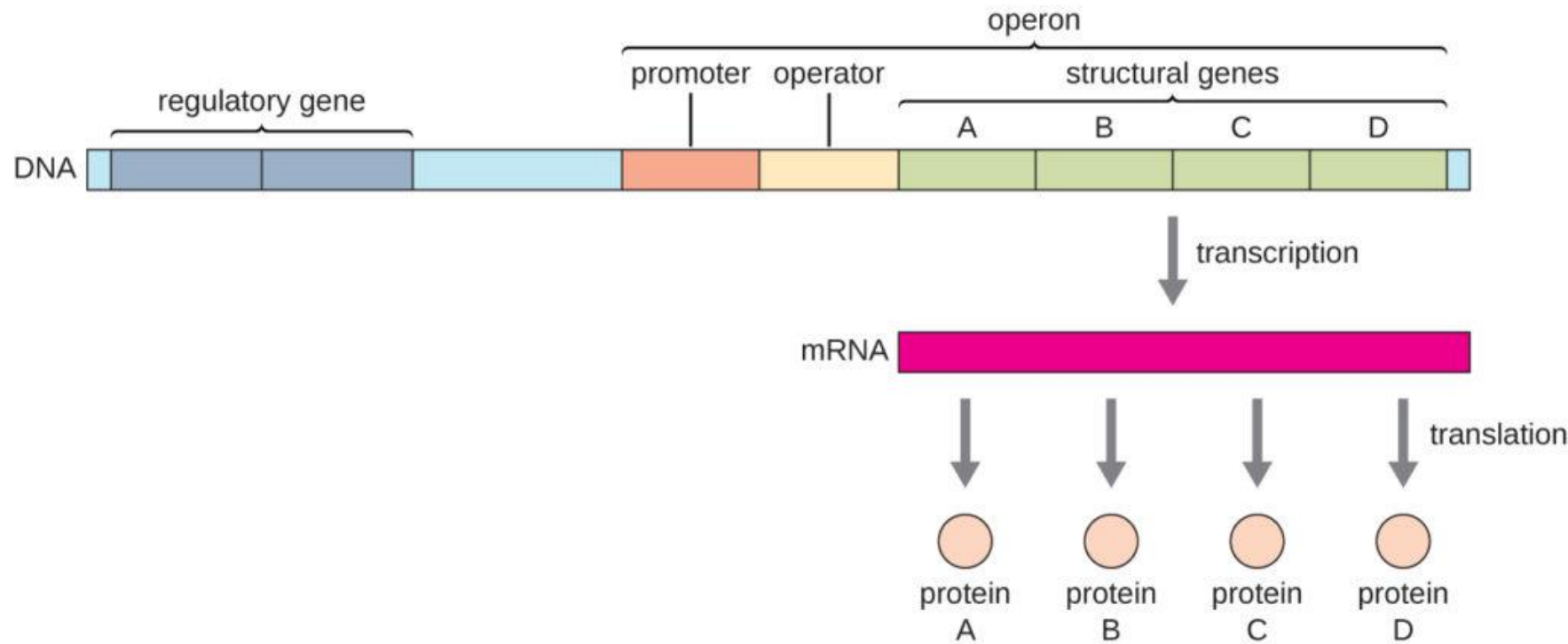
- Many **polycistronic**, *i.e.* many genes organised in an operon, but also **monocistronic**

Eukaryotic gene expression



- Usually monocistronic gene structure with exons and introns
- Eukaryotic mRNAs are poly-adenylated
- Pre-mRNA undergoes splicing to remove introns leaving exons that encode a protein (note there is also alternative splicing)

Prokaryotic gene expression



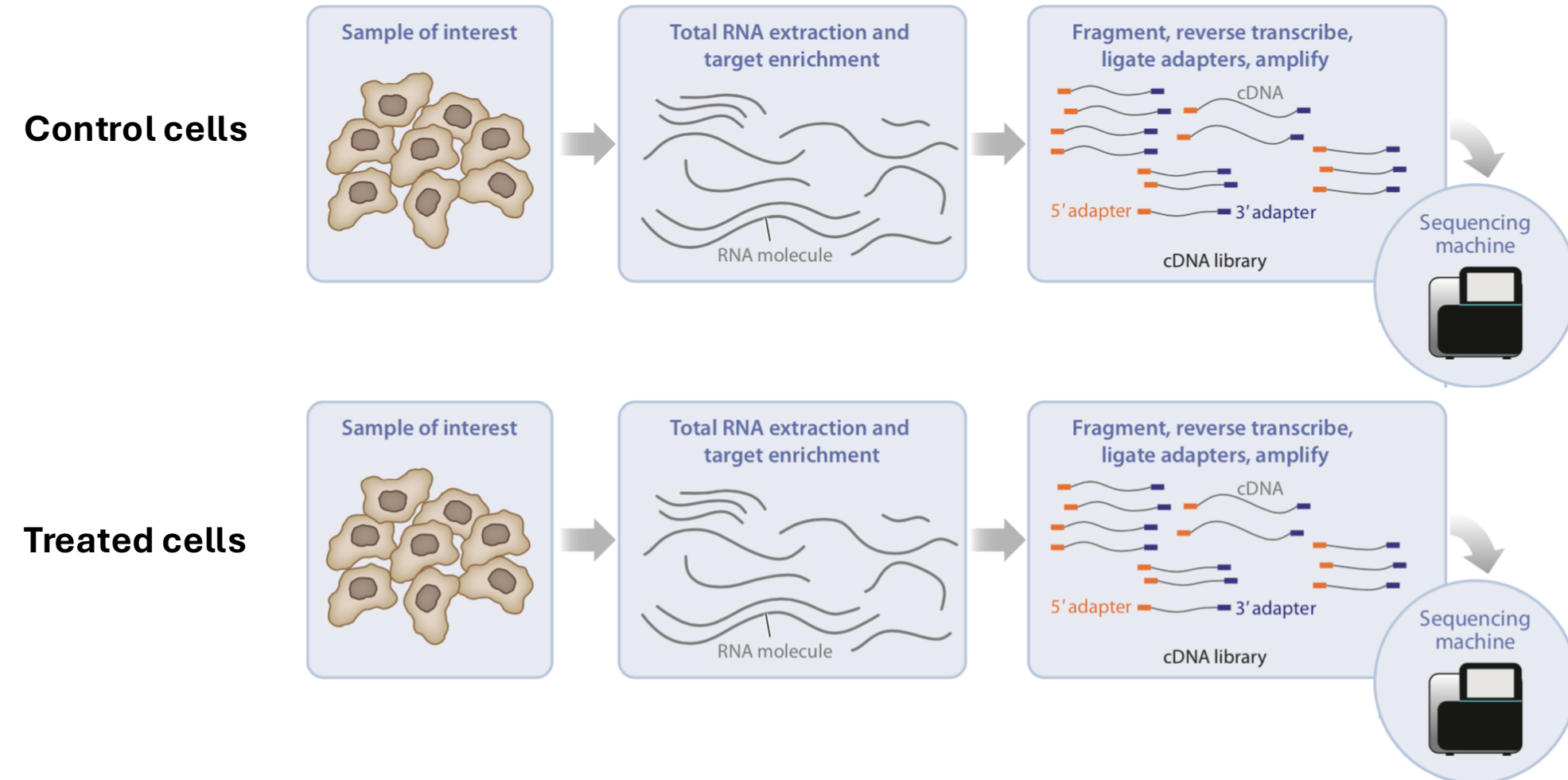
- Polycistronic mRNA encodes many open reading frames encoding individual proteins

Transcriptome analysis using RNA sequencing (RNAseq)

- **Short read sequencing (today's topic)**
 - **Illumina (50 – 300 base pairs)**
- Long-read sequencing
 - 1000 – 10000 base pairs
 - Oxford Nanopore
 - PacBio

Illumina sequencing technology – RNAseq

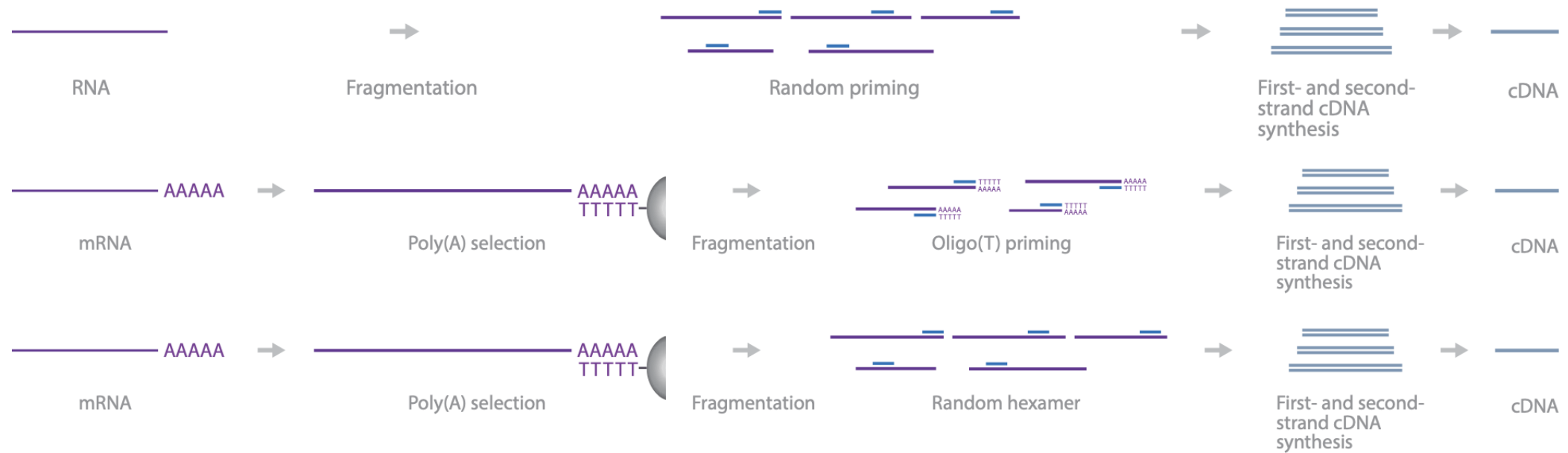
- No direct RNA sequencing! Instead, RNA is enzymatically reverse transcribed to yield cDNA that is sequenced



- cDNA (copyDNA) libraries usually differ in size, *i.e.* in the total number of sequenced reads, between samples
- **Impact on data analysis:** normalization by library size needed for inter-sample comparison

Illumina sequencing technology – RNAseq

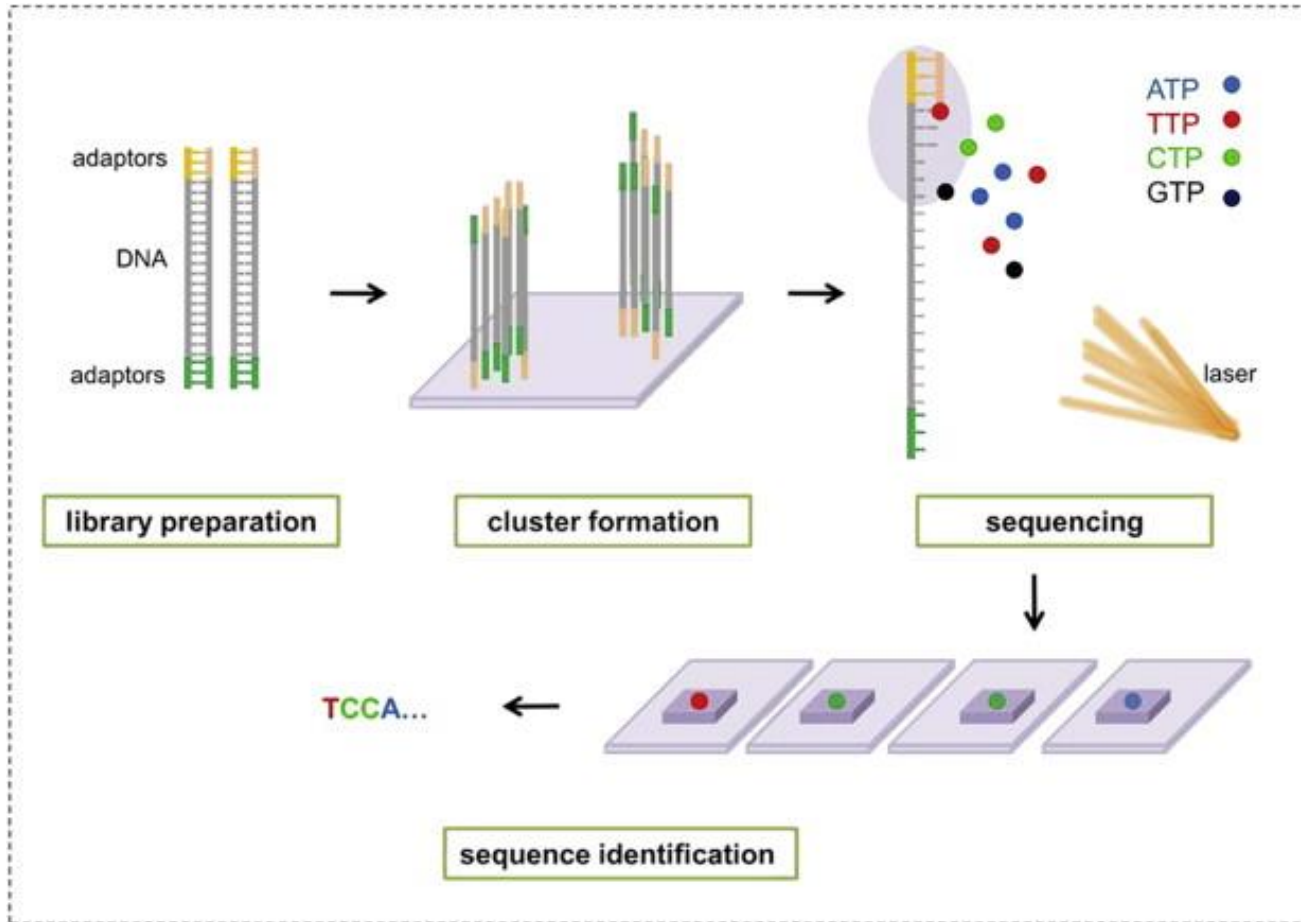
- Random or oligo(T) priming for reverse transcription
 - For prokaryotes: random primers
- Enrichment of poly-adenylated RNAs possible (target enrichment)



A schematic overview of RNA-Seq.

Illumina sequencing technology

- Massively parallel sequencing of DNA immobilized on a solid surface of a flow cell

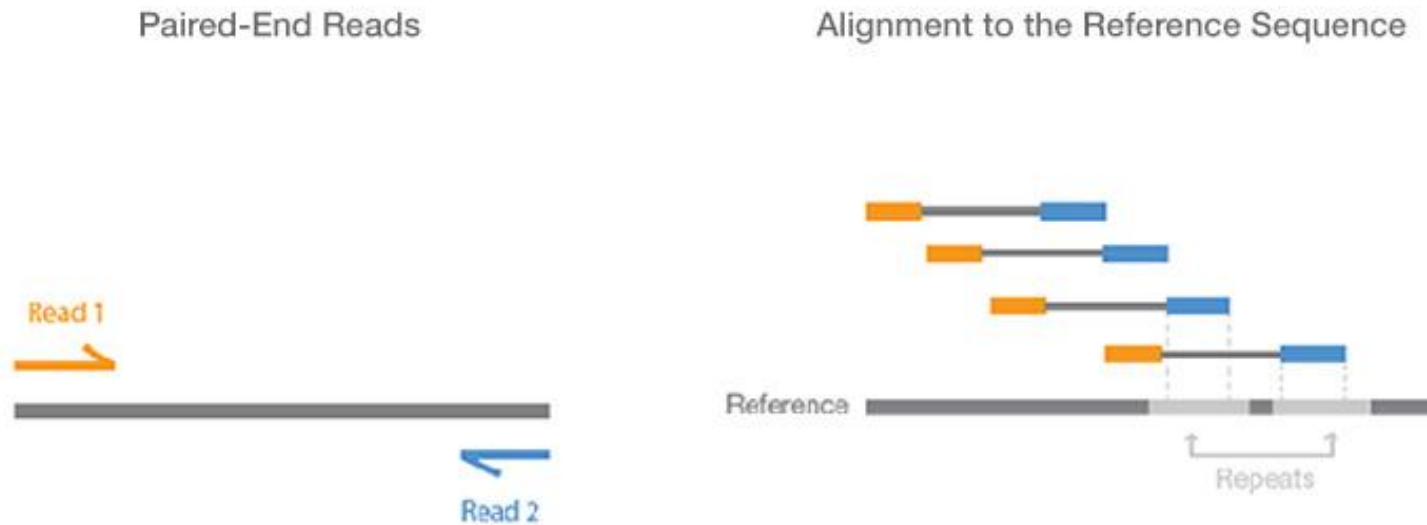


- Adaptors are ligated to fragmented DNA
- DNA is attached on a surface and amplified forming clusters
- Sequencing-By-Synthesis (SBS):
 - Incorporation of fluorophore-labeled dNTPs is measured during DNA synthesis
- Impact on data analysis:** adaptor sequences have to be removed prior to analysis (adaptor trimming)

Very good video with further details on Illumina paired-end sequencing (found by Albert and presented in metagenomics course):

<https://www.youtube.com/watch?v=HMyCqWhwB8E>

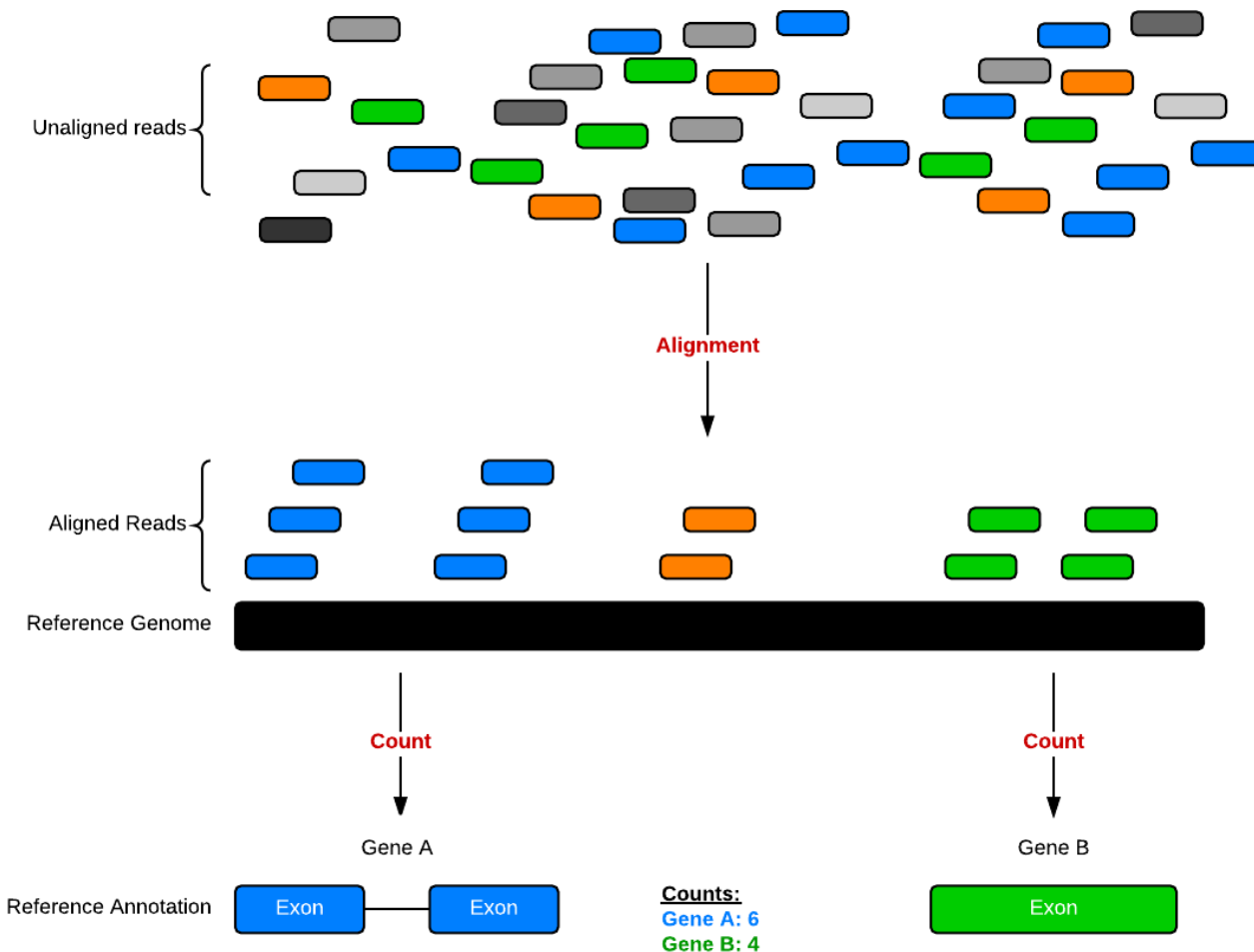
Illumina sequencing technology – Paired-end sequencing



- Higher accuracy when DNA is sequenced from both ends
- Reads can be aligned with higher precision, especially in complex genomic regions (repetitive sequences)
- Detection of genomic rearrangements
- Detection of gene fusions and novel transcripts

Paired-end sequencing enables both ends of the DNA fragment to be sequenced. Because the distance between each paired read is known, alignment algorithms can use this information to map the reads over repetitive regions more precisely. This results in much better alignment of the reads, especially across difficult-to-sequence, repetitive regions of the genome.

Major steps and goals in RNAseq



Major steps in RNAseq analysis

- Quality control (QC)
- Alignment: map reads to a reference sequence (e.g. genome)
- Read quantification: count/quantify how many reads are aligned to a specific genomic region (feature)

Focus in this course is quantification

- One major goal of RNAseq is differential expression (DE) analysis comparing read counts of a specific genomic feature between biological experiments

Control cells

Gene

==

Treated cells

Gene

====

Further goals (not covered today)

- New transcript detection
- New isoform detection (alternative splicing)

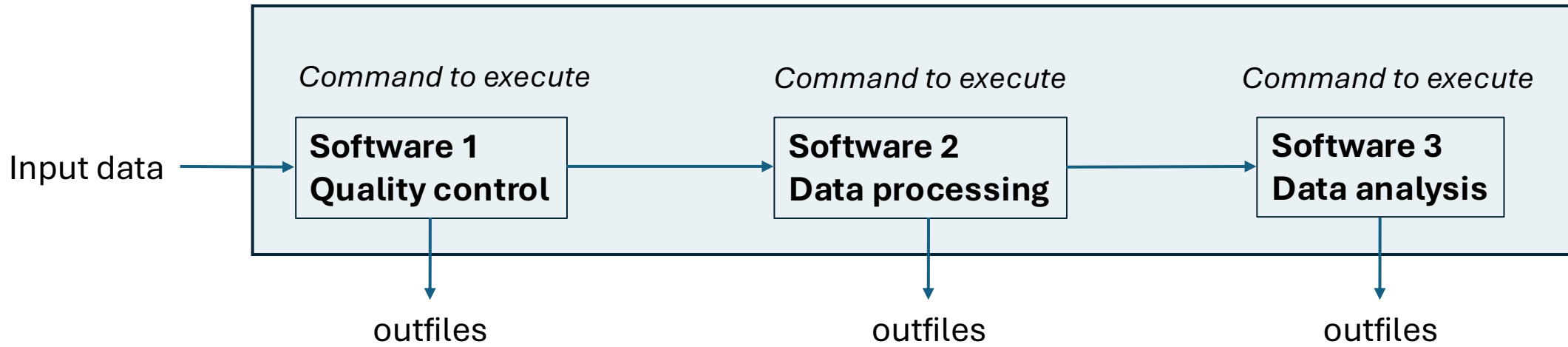
Bioinformatics for Illumina short read data

- **1. Data processing (nf-core/rnaseq pipeline) – selected software tools**
 - Quality control (QC), MultiQC
 - Read alignment
 - Read quantification
- **2. Post-processing data analysis**
 - Normalization methods
 - Differential gene expression (DGE *or* DE) analysis
 - Statistics
 - DESeq2 (R programming language)
 - Functional analysis for biological interpretation
 - Overrepresentation analysis (ORA)
 - Gene-set enrichment analysis (GSEA)

1. Data processing – pipelining with nextflow

- RNAseq data analysis requires the consecutive execution of individual software tools
- Executing software tools one by one using the command line is possible but requires many manual steps
- **Reproducible data processing and analysis** can be achieved by chaining individual software versions (**containerized using e.g. docker**) using a **dedicated pipelining tool, e.g. nextflow**
- Execution of pipeline is done using a single command
- Nextflow: Portable (across different platforms), scalable (parallel data processing), reproducible (standardized workflow)

Pipeline: Single command to run the pipeline



1. Data processing – nf-core



The nf-core framework for community-curated bioinformatics pipelines

[Philip A. Ewels](#), [Alexander Peltzer](#), [Sven Fillinger](#), [Harshil Patel](#), [Johannes Alneberg](#), [Andreas Wilm](#),
[Maxime Ulysse Garcia](#), [Paolo Di Tommaso](#) & [Sven Nahnsen](#) 

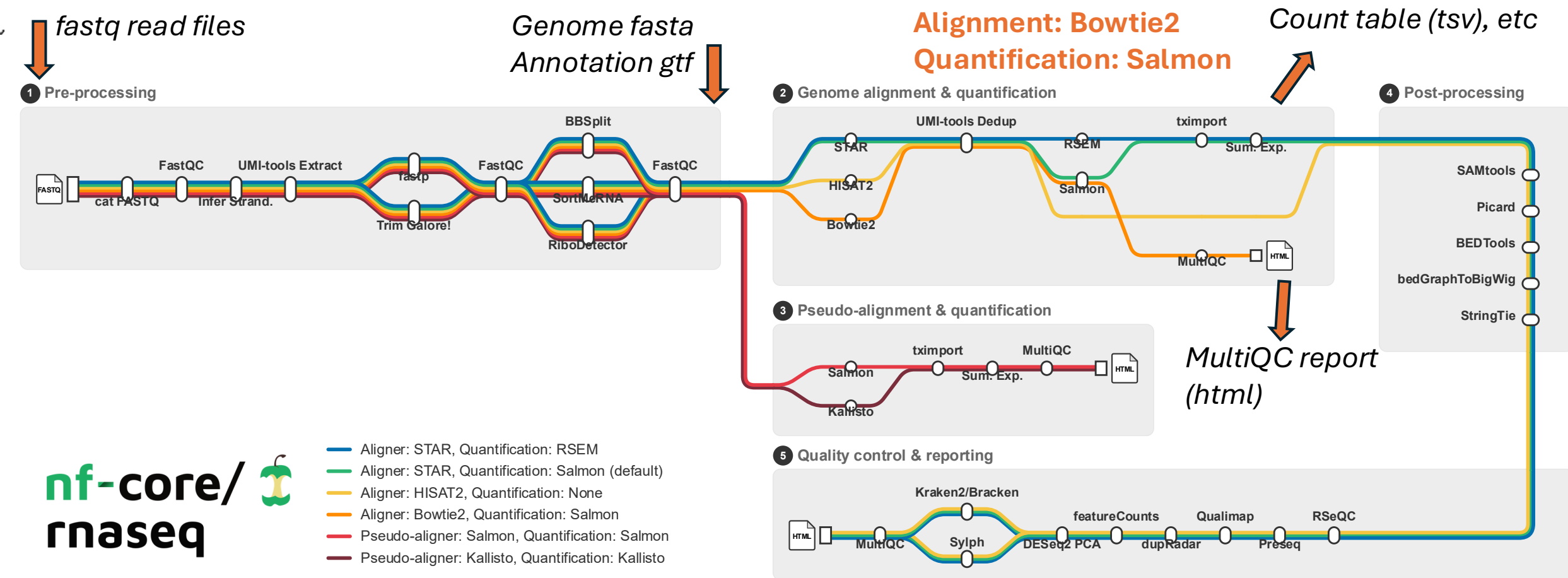
[Nature Biotechnology](#) **38**, 276–278 (2020) | [Cite this article](#)

34k Accesses | **166** Altmetric | [Metrics](#)

- **Collection of nextflow pipelines for bioinformatics**
 - 150 pipelines available: 91 released, 46 under development, 13 archived (status 19 May 2026)
 - Standardized and parallel data processing (e.g. in the cloud)
 - Pipelines undergo rigorous testing before release
 - Continuously tested and maintained pipelines
 - Direction of development is heavily discussed with a community of developers
 - Active community of developers and users with fast support and troubleshooting
 - Regular world-wide hackathons

1. Data processing – nf-core/rnaseq pipeline

- We will follow the **orange** route to process bacterial RNAseq data



created with nf-metro v0.5.4

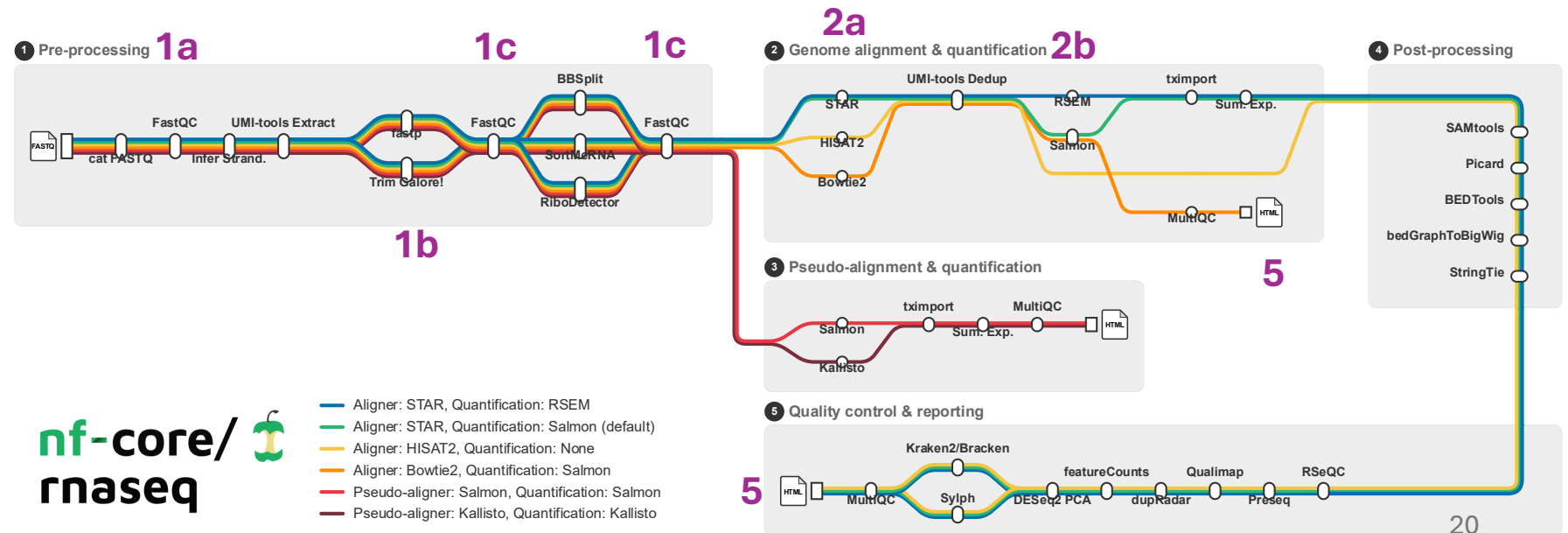
1. Data processing – nf-core/rnaseq pipeline

Overview of selected individual steps

- 1a) **Quality control (QC)** with **FastQC** of sequencing reads
- 1b) **Adaptor trimming** with **Trim Galore** to remove adaptor sequences that were ligated to DNA prior to sequencing
- 1c) **FastQC** again
- 2a) Read alignment with **Bowtie2** (or other aligners like STAR for wukaryotes) to map of reads against a reference sequence (e.g. genome sequence)
- 2b) Read quantification with **Salmon** to quantify how many reads were mapped to a specific genomic feature (e.g. gene)
- ...
- 5) **MultiQC** report consolidates all quality control results obtained from several softwares

Input files

- Sequencing files (.fastq)
- Genome sequence (.fasta)
- Genome annotation (.gtf)
- (Optional additional files)



1. Data processing input– sequencing files (.fastq)

- Fastq files contain information about the sequenced reads (one example entry shown below)
- One entry has 4 lines

[illegible]

Line 1: A sequence identifier with information about the sequencing run and the cluster.

Line 2: DNA sequence (the base calls; A, C, T, G and N).

Line 3: A separator, a plus (+) sign.

Line 4: Base call quality scores (Phred +33 encoded, using ASCII characters to represent the numerical quality scores).

1. Data processing input – genome sequence file (.fasta, .fna)

- Usually gzipped files to compress file size (.fna.gz, .fasta.gz)
- DNA sequence for each chromosome as fasta entry (starting with “>”)
- For our *E. coli* case study, there is only one fasta entry representing the entire genome sequence of the single chromosome (example shown below)
- For Eukaryotes, each fasta entry has a header and the DNA sequence of the respective chromosome

>NC_000913.3 Escherichia coli str. K-12 substr. MG1655, complete genome

```
AGCTTTTCATTCTGACTGCAACGGGCAATATGTCTCTGTGTGGATTAAAAAAAGAGTGTCTGATAGCAGCTTCTGAACTG
GTTACCTGCCGTGAGTAAATTAAAATTTTATTGACTTAGGTCACTAAATACTTTAACCAATATAGGCATAGCGCACAGAC
AGATAAAAATTACAGAGTACACAACATCCATGAAACGCATTAGCACCACCATTACCACCACCATCACCATTACCACAGGT
AACGGTGCGGGGCTGACGCGTACAGGAAACACAGAAAAAAGCCCGCACCTGACAGTGCGGGGCTTTTTTTTTTCGACCAA
GGTAACGAGGTAACAACCATGCGAGTGTTGAAGTTCGGCGGTACATCAGTGGCAAATGCAGAACGTTTTCTGCGTGTTGC
CGATATTCTGGAAAGCAATGCCAGGCAGGGGCGAGGTGGCCACCGTCCTCTCTGCCCCCGCCAAAATCACCAACCACC
TGGT...
```

1. Data processing input – annotation file (.gtf)

- General Transfer Format (GTF) with nine columns (snapshot of *E. coli* MG1655 gtf file shown below)
- Contains information about annotated sequence features (e.g. gene, CDS, exon, transcript, etc)

#gtf-version 2.2								
#!genome-build ASM584v2								
#!genome-build-accession NCBI_Assembly:GCF_000005845.2								
NC_000913.3	RefSeq	gene	190	255	.	+	.	gene_id "b0001"; transcript_id ""; db_xref "ASAP:ABE-0000006"; db_xref "ECOCYC:EG11277"; db_xref "GeneID:944742"; gbkey "gene";
NC_000913.3	RefSeq	CDS	190	252	.	+	0	gene_id "b0001"; transcript_id "unassigned_transcript_1"; db_xref "UniProtKB/Swiss-Prot:P0AD86"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	start_codon	190	192	.	+	0	gene_id "b0001"; transcript_id "unassigned_transcript_1"; db_xref "UniProtKB/Swiss-Prot:P0AD86"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	stop_codon	253	255	.	+	0	gene_id "b0001"; transcript_id "unassigned_transcript_1"; db_xref "UniProtKB/Swiss-Prot:P0AD86"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	gene	337	2799	.	+	.	gene_id "b0002"; transcript_id ""; db_xref "ASAP:ABE-0000008"; db_xref "ECOCYC:EG10998"; db_xref "GeneID:945803"; gbkey "gene";
NC_000913.3	RefSeq	CDS	337	2796	.	+	0	gene_id "b0002"; transcript_id "unassigned_transcript_2"; db_xref "UniProtKB/Swiss-Prot:P00561"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	start_codon	337	339	.	+	0	gene_id "b0002"; transcript_id "unassigned_transcript_2"; db_xref "UniProtKB/Swiss-Prot:P00561"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	stop_codon	2797	2799	.	+	0	gene_id "b0002"; transcript_id "unassigned_transcript_2"; db_xref "UniProtKB/Swiss-Prot:P00561"; db_xref "GenBank:NP_417442.1";
NC_000913.3	RefSeq	gene	2801	3733	.	+	.	gene_id "b0003"; transcript_id ""; db_xref "ASAP:ABE-0000010"; db_xref "ECOCYC:EG10999"; db_xref "GeneID:947498"; gbkey "gene";
NC_000913.3	RefSeq	transcript	3648063	3648144	.	+	.	gene_id "b4712"; transcript_id "unassigned_transcript_3591"; db_xref "ECOCYC:G0-10751"; db_xref "GeneID:19351531"; gbkey "transcript";
NC_000913.3	RefSeq	exon	3648063	3648144	.	+	.	gene_id "b4712"; transcript_id "unassigned_transcript_3591"; db_xref "ECOCYC:G0-10751"; db_xref "GeneID:19351531"; gbkey "exon";
NC_000913.3	RefSeq	gene	3648294	3648375	.	+	.	gene_id "b4713"; transcript_id ""; db_xref "ECOCYC:G0-10752"; db_xref "GeneID:19351532"; gbkey "gene"; gene "agrB"; gene "agrC";
NC_000913.3	RefSeq	transcript	3648294	3648375	.	+	.	gene_id "b4713"; transcript_id "unassigned_transcript_3592"; db_xref "ECOCYC:G0-10752"; db_xref "GeneID:19351532"; gbkey "transcript";
1	2	3	4	5	6	7	8	9

- 1. seqname:** name of the chromosome or scaffold
- 2. source:** name of the program that generated this feature, or the data source (database or project name)
- 3. feature:** feature type name, e.g. gene, exon
- 4. start:** start position* of the feature, with sequence numbering starting at 1.
- 5. end:** end position* of the feature, with sequence numbering starting at 1.
- 6. score:** a floating point value.
- 7. strand:** defined as + (forward) or - (reverse).
- 8. frame:** one of '0', '1' or '2'. '0' indicates that the first base of the feature is the first base of a codon, '1' that the second base is the first base of a codon, and so on..
- 9. attribute:** a semicolon-separated list of tag-value pairs, providing additional information about each feature.

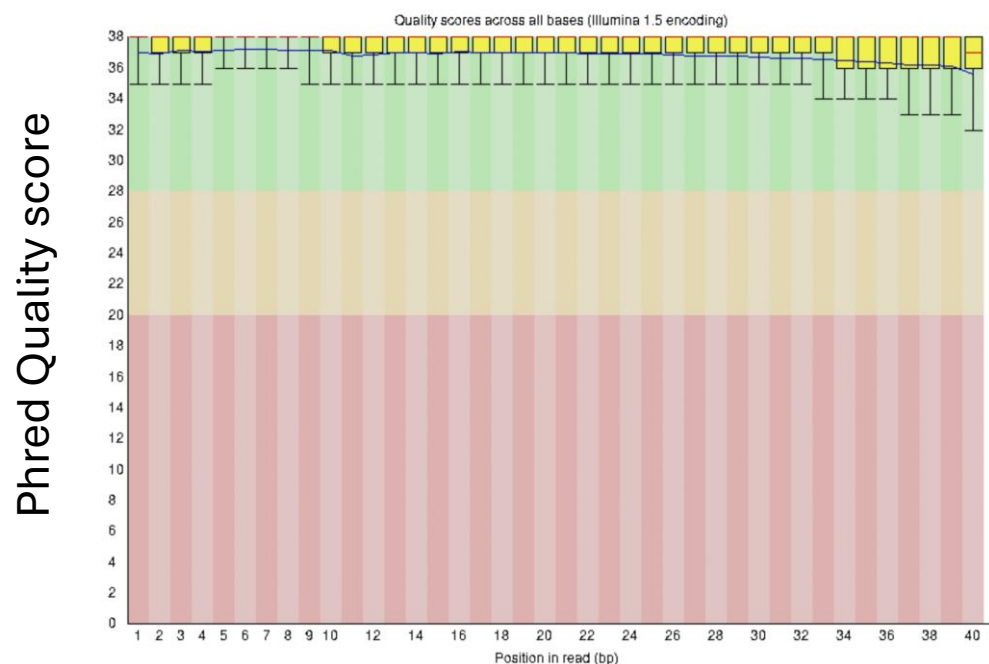
**We will align reads
against the feature
“CDS”**

1. Data processing – Quality control with FastQC

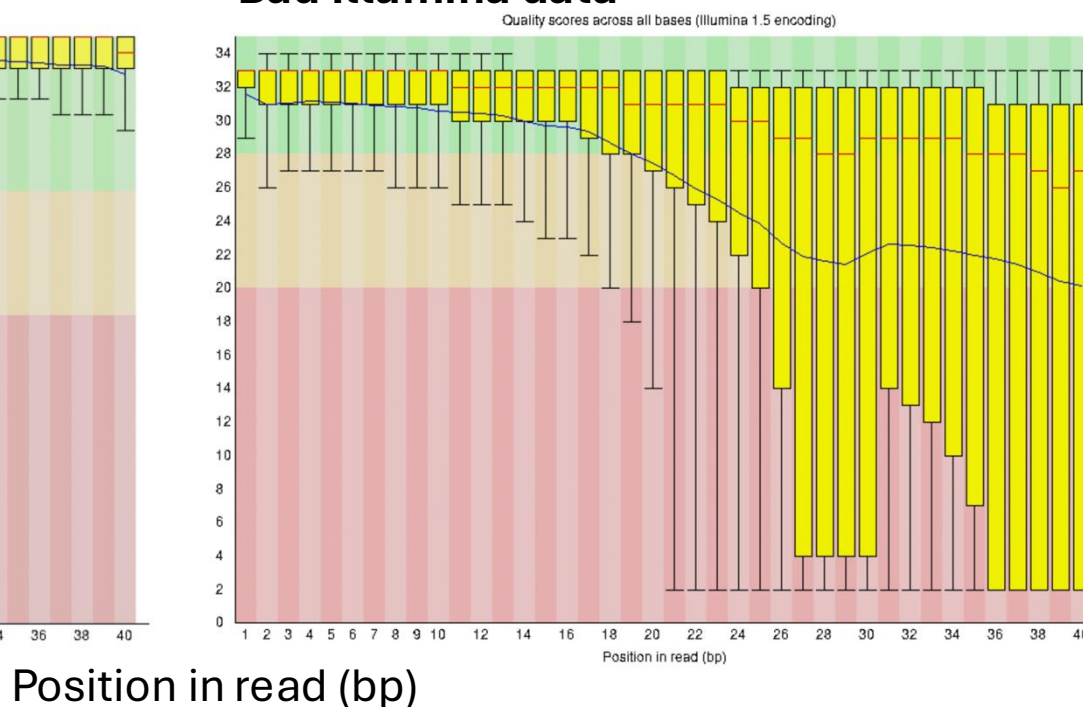
- FastQC assesses the quality per base pair position of sequencing reads and assigns a Phred quality score (illustrated below)
- Assesses also read length distribution as well as per sequence GC, N and adaptor content
- Conducted twice: before and after adapter trimming

Phred Quality Score	Probability of incorrect base call	Base call accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1000	99.9%
40	1 in 10,000	99.99%

Good Illumina data



Bad Illumina data



1. Data processing - Read alignment with Bowtie2 or STAR

Reference sequence

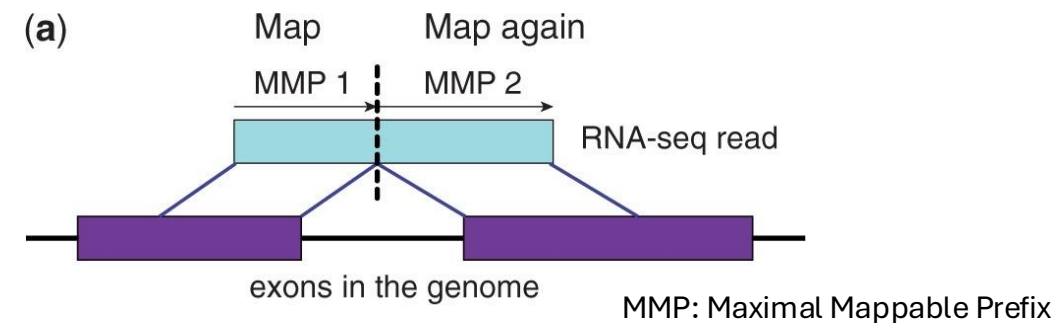
Bowtie2

- Successor of the Bowtie aligner
- Alignment with gaps
- Alignment against genome or transcriptome
- Splicing-Unaware: NO *de novo* detection of splice junctions (annotation of splicing events required)
- Good choice for prokaryotic data; can be used for eukaryotic data

STAR

- Spliced Transcripts Alignment to a Reference (STAR)
- Very fast aligner
- Alignment against genome or transcriptome
- Splicing-aware: *de novo* detection of splice junctions (no prior annotation of splicing events required)
- Good choice for eukaryotic data; can in principle be used for prokaryotic data with specific parameter settings but issues observed using previous versions of the nf-core/rnaseq pipeline

***De novo* detection of splice junctions with STAR**



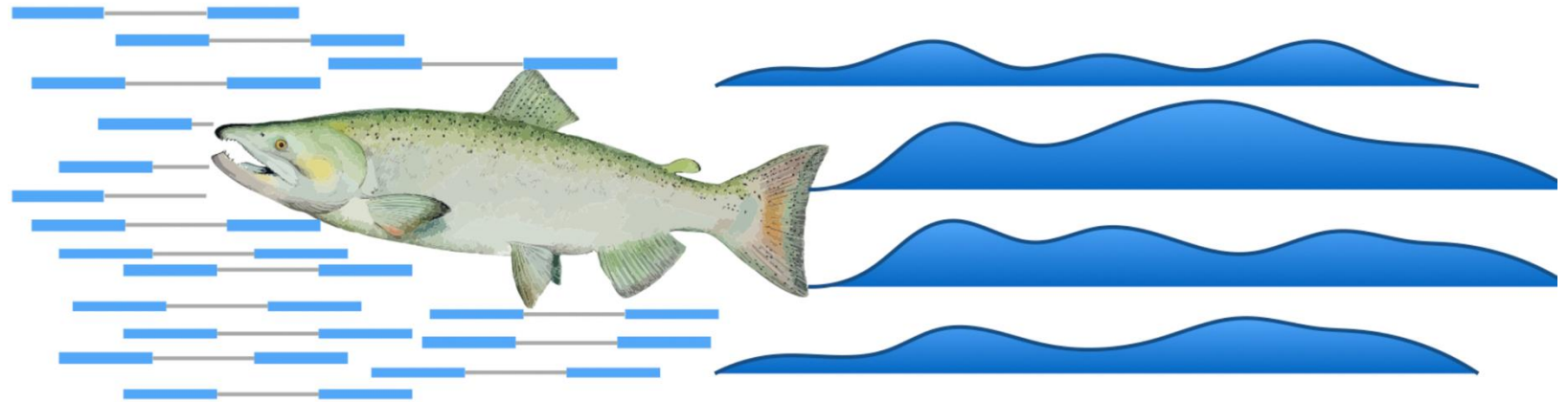
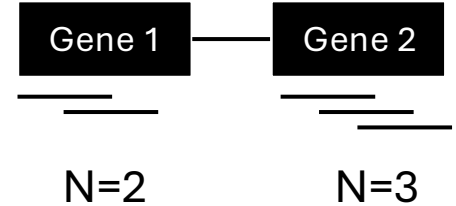
Bowtie2 paper: <https://doi.org/10.1038/nmeth.1923>

Bowie paper: <https://doi.org/10.1186/gb-2009-10-3-r25>

STAR paper: <https://doi.org/10.1093/bioinformatics/bts635>

1. Data processing - Read quantification with Salmon

- More complex quantification procedure compared to other read counters/quantifiers like featureCounts
- Handles counting of multi-mapping reads
- More accurate quantification of reads achieved by considering sample-specific parameters and biases of RNAseq data:
 - positional biases in coverage
 - sequence-specific biases at the 5' and 3' ends of sequenced fragments
 - fragment-level GC bias
 - strand-specific protocols
 - fragment length distribution



Hands-on data analysis: case study background

Article



SOURCE
DATA



TRANSPARENT
PROCESS



OPEN
ACCESS



CHECK FOR
UPDATES

EMBO
Molecular Medicine

Saccharin disrupts bacterial cell envelope stability and interferes with DNA replication dynamics

Rubén de Dios ¹, Kavita Gadar ¹, Chris R Proctor¹, Evgenia Maslova¹, Jie Han ¹, Mohamed A N Soliman ¹, Dominika Krawiel¹, Emma L Dunbar², Bhupender Singh³, Stelinda Peros⁴, Tom Killelea⁵, Anna-Luisa Warnke ⁶, Marius M Haugland⁶, Edward L Bolt⁵, Christian S Lentz³, Christian J Rudolph ⁴ & Ronan R McCarthy ¹✉

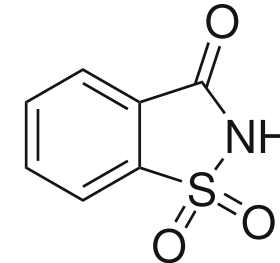
de Dios, R., Gadar, K., Proctor, C.R. et al.

EMBO Mol Med **17**, 993–1017 (2025).

<https://doi.org/10.1038/s44321-025-00219-1>

Data set link:

<https://www.ebi.ac.uk/ena/browser/view/PRJNA1158806>



Background

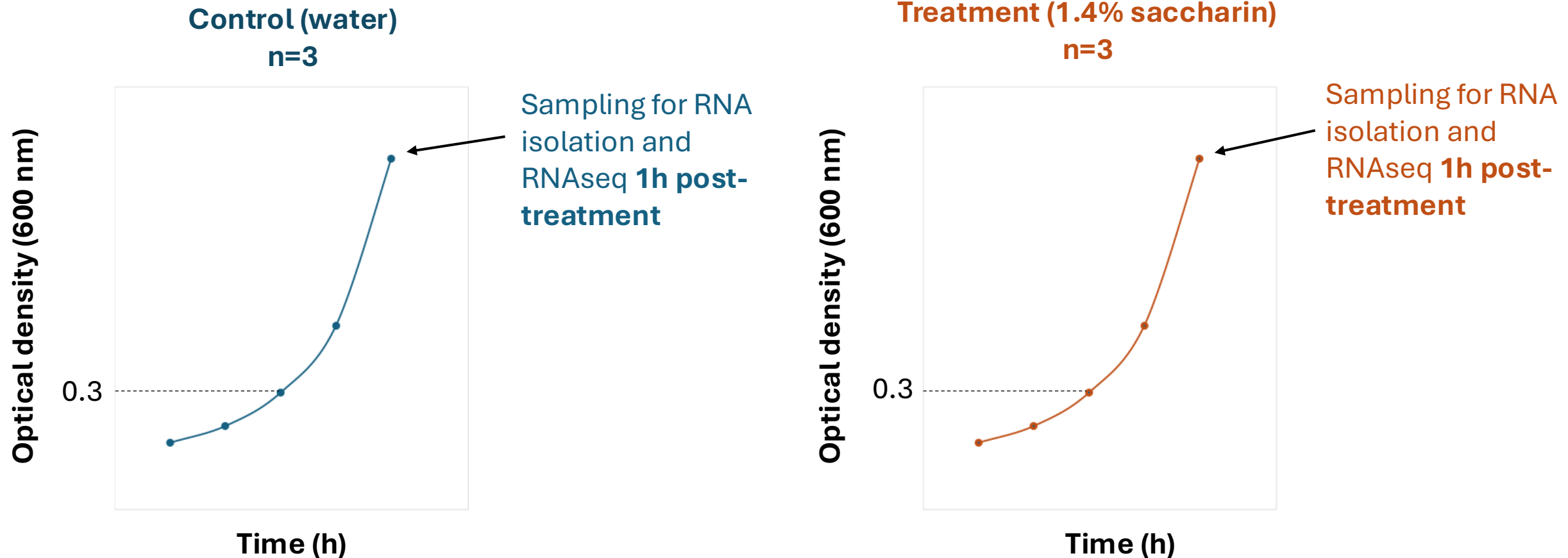
- Saccharin (benzosulfimide, E954) is an artificial sweetener used in human diet for > 100 years
- Increasing evidence that saccharin has an effect on the gut microbiome, however, precise mechanisms unknown
- Various bacteria studied using a number of experimental assays including RNAseq of *E. coli* MG1655

Key findings

- Saccharin can
 - inhibit the growth of both Gram-negative and Gram-positive bacteria
 - disrupt key phenotypes linked to host colonisation, such as motility and biofilm formation

Hands-on data analysis: case study design

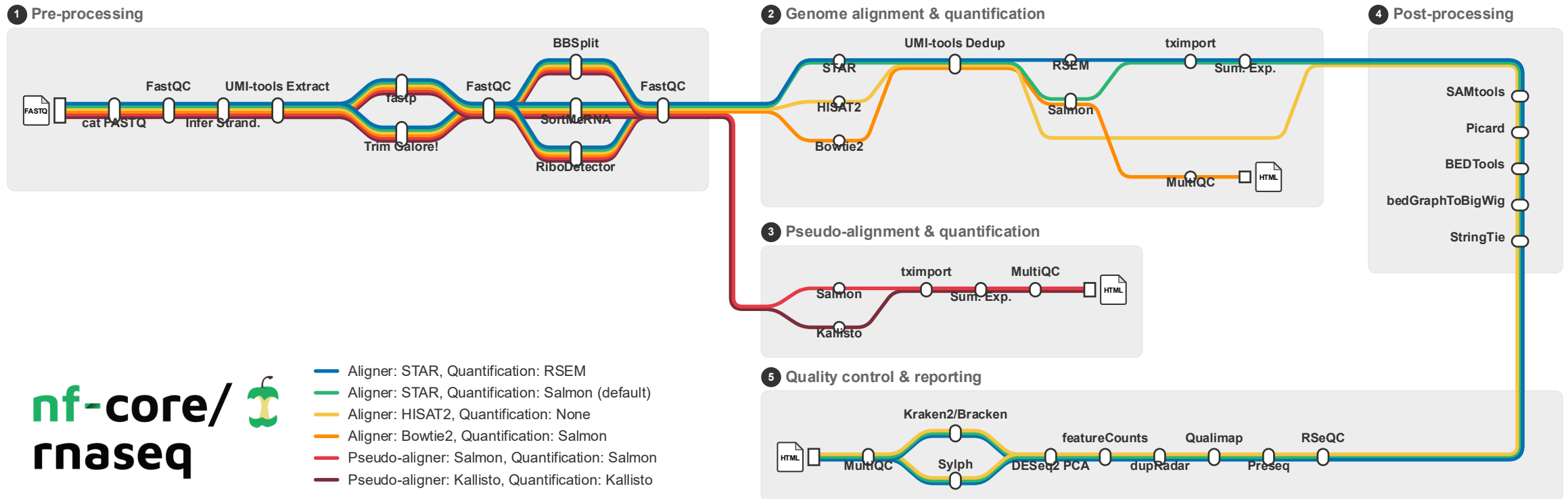
- *E. coli* MG1655 was cultivated in LB medium until early exponential phase (OD600=0.3)
- At this point, 1.4% saccharin (treatment) or water (control) was added
- After an additional hour of cultivation samples were taken, RNA isolated and sequenced
- Biological triplicates per condition



Growth curves are schematic, *i.e.* they were created for illustration purposes (not taken from the paper)

Hands-on data processing with nf-core/rnaseq

- Processing the entire data set would need more computational resources
- We will process 2 size-reduced samples of the data set in the codespace instead
 - Instructions can be found here: https://biosustain.github.io/dsp_transcriptomics_training/instructions.html



created with nf-metro v0.5.4

Lunch break

1. Data processing – output files

- nf-core pipeline outputs from the different software tools are saved in dedicated subfolders

- **Multi-QC report (html) examples:**

- General stats
 - STAR alignment stats
- FastQC(raw)
 - Sequence Quality Histograms
 - Per Sequence GC Content
 - Per Base N Content
- Software versions
- nf-core/rnaseq Workflow Summary

- **Bowtie2_salmon subfolder:**

- Count tables: read counts per sequence feature

Counts results
for downstream
analysis

Overall subfolder structure

```
.  
├── bowtie2_salmon  
├── fastqc  
├── fq_lint  
├── multiqc  
├── pipeline_info  
└── trimalore
```

Multiqc subfolder

Multi-QC report
for inspection

```
├── multiqc  
│   └── bowtie2_salmon  
│       ├── multiqc_report_data  
│       ├── multiqc_report.html  
│       └── multiqc_report_plots
```

Bowtie2-Salmon subfolder (partial)

```
├── saccharin_1.markdup.sorted.bam  
├── saccharin_1.markdup.sorted.bam.bai  
├── salmon.merged.gene_counts_length_scaled.tsv  
├── salmon.merged.gene_counts_scaled.tsv  
├── salmon.merged.gene_counts.tsv  
├── salmon.merged.gene_lengths.tsv  
├── salmon.merged.gene.SummarizedExperiment.rds  
├── salmon.merged.gene_tpm.tsv  
├── salmon.merged.transcript_counts.tsv  
└── salmon.merged.transcript_lengths.tsv
```

2. Post-processing data analysis

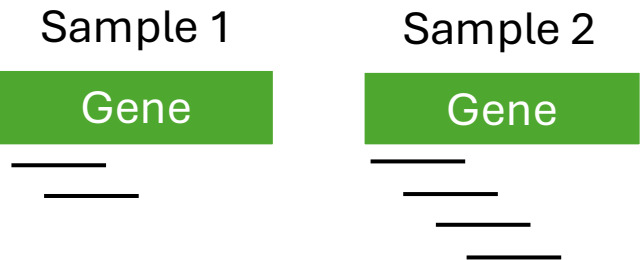
- Normalization methods
- Differential gene expression (DGE *or* DE) analysis
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 - DESeq2 (R language)
- Functional analysis for biological interpretation
 - Over-representation analysis (ORA)
 - Gene-set enrichment analysis (GSEA)

Normalization methods

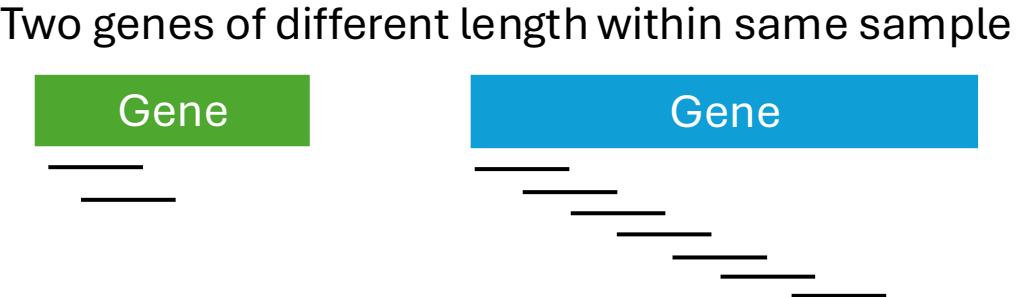
Read counts per gene depend on (among others):

- 1. Library size (count of total reads)
- 2. Gene length
- 3. RNA composition

1. Library size



2. Gene length

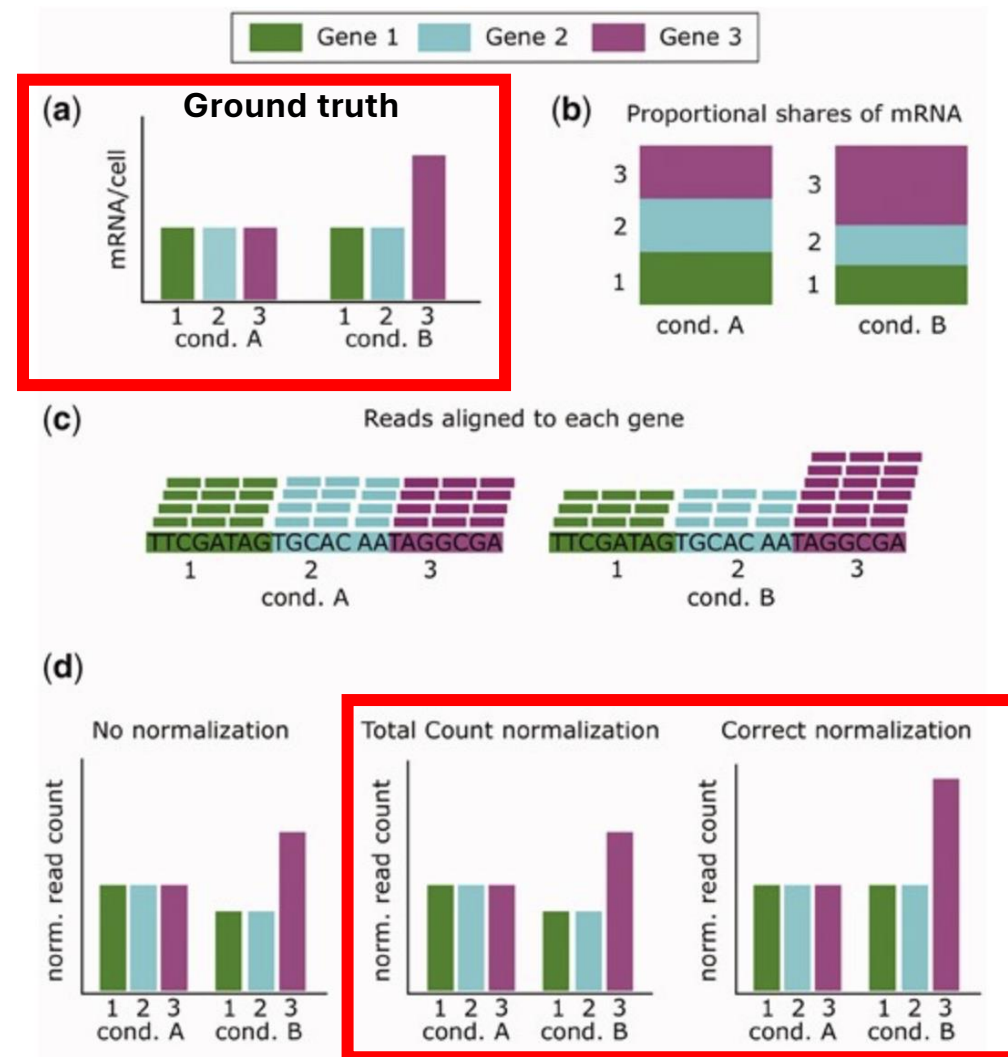
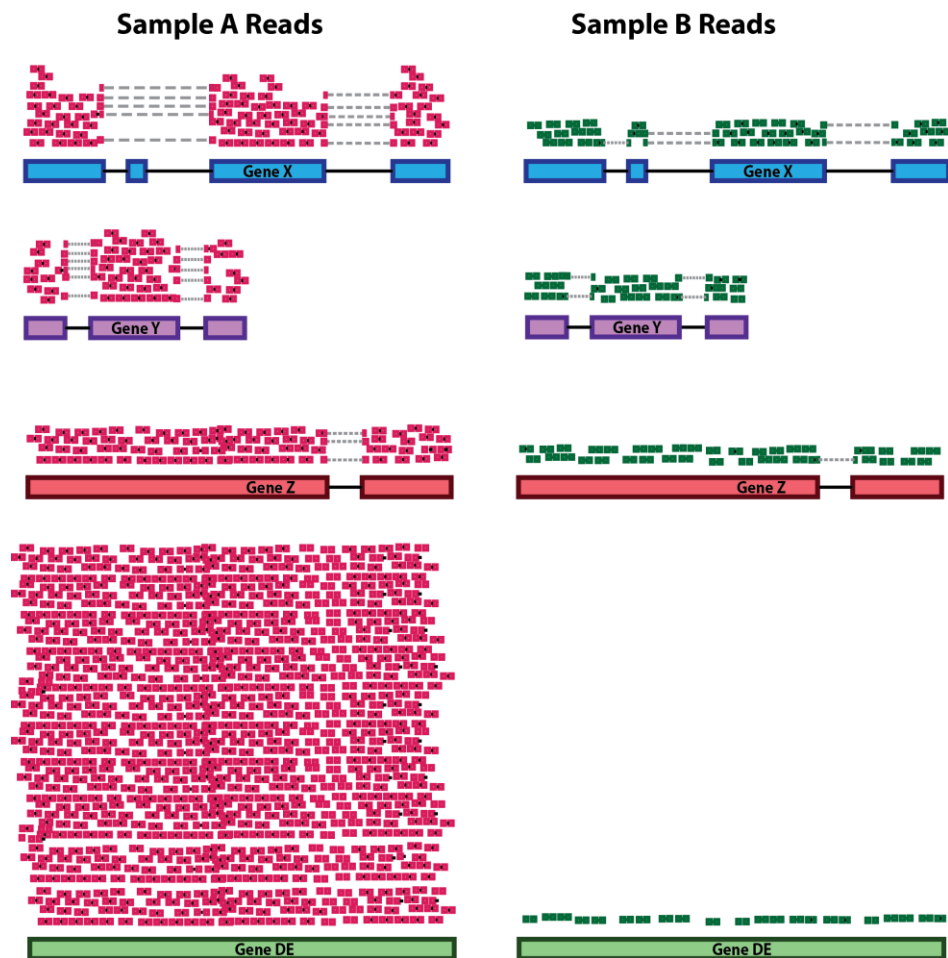


Read count	2	4	2	8
Count of total reads	n	2n	n	n
Gene length	L	L	L	4L
Normalized count	2	2	2	2
Relevant for	Inter-sample comparison		Intra-sample comparison	

Normalization methods

Read counts per gene depend on (among others):

3. RNA composition



- Many reads for a single highly expressed gene can make genes expressed at lower levels appear to be even lower
- This doesn't reflect the ground truth (see figure a)

Normalization methods – selection

Method	Description	Library size normalized	Gene length normalized	RNA composition normalized	Useful for comparisons	Comment	Refs
CPM	Counts per million mapped reads	yes	no	no	Inter-sample of same sample group		
RPKM/FPKM	Reads/fragments per kilobase per million mapped reads	yes	yes	no	Intra-sample	Not so much in use anymore	
TPM	Transcripts per million	yes	yes	no	Intra-sample; inter-sample of same sample group	Sum of gene TPM values is the same across samples	Link
TMM	Trimmed mean of M values	yes	no	yes	Inter-sample and quantitative DE analysis	Used in edgeR package	Link
DEseqCounts	Median of the ratios	yes	no	yes	Inter-sample and quantitative DE analysis	Used in DESeq2 package	Link

- Choice of a normalization method can depend on several factors, e.g. the downstream analysis
- DESeq2 is one of the standards for quantitative estimation of differential gene expression

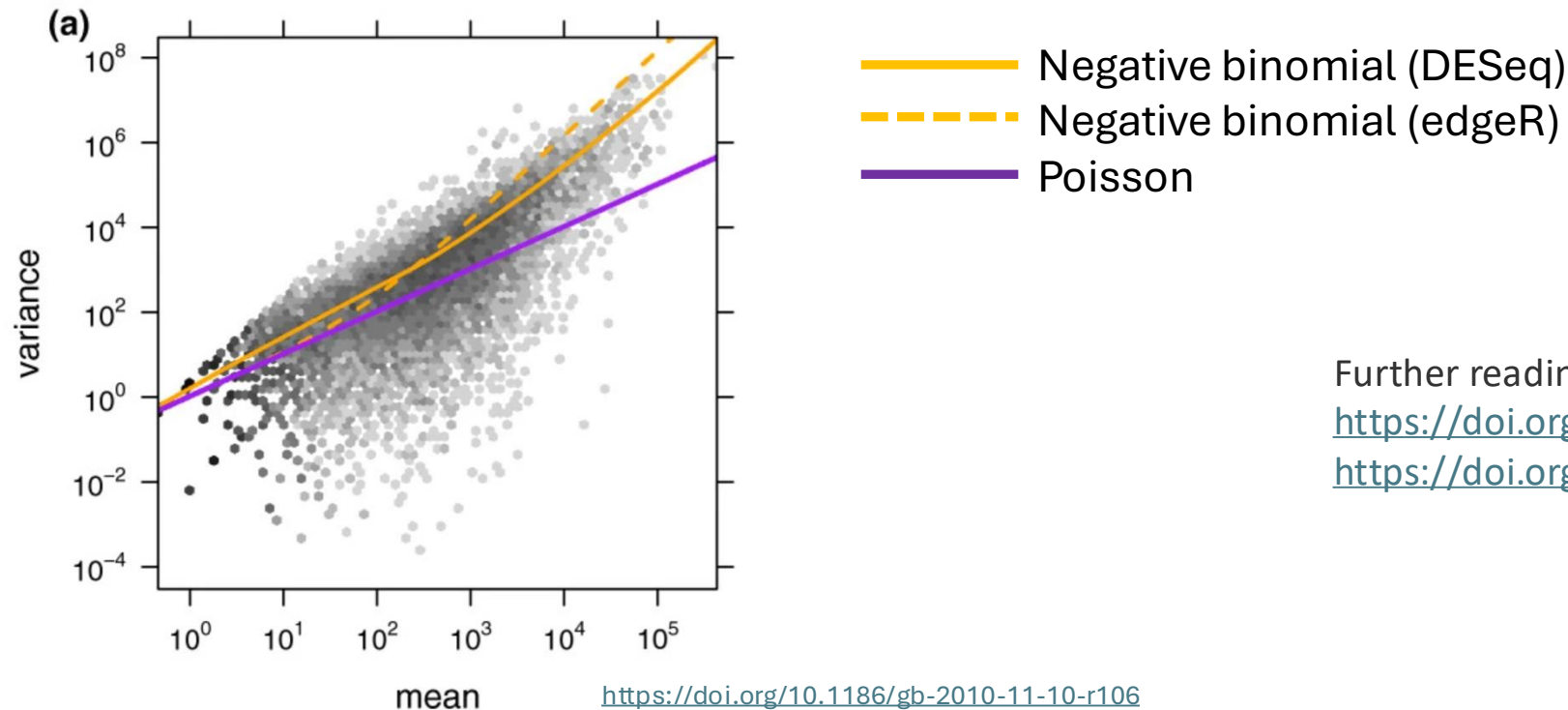
[Source link](#)

Differential gene expression analysis with DESeq2

- Main goal of differential gene expression analysis is to determine genes that are significantly differentially expressed between experimental conditions
 - Effect size: calculation of **fold-change** or **log2-fold-change (LFC)**
 - Statistical inference: calculation of **p-values adjusted for multiple testing**
- DESeq2 and its predecessor DESeq are R libraries specifically designed for the statistical analysis of bulk RNAseq count data
 - **Important:** never use normalized reads as input to DESeq2!!!
 - DESeq2 accepts only raw read counts and uses the on the previous slide described internal count normalization!!!
- Statistical model of DESeq2
 - RNAseq data distribution
 - Normalization (DESeqCounts)
- Hands-on

Differential gene expression analysis – data distribution

- RNAseq read counts have to be modelled by a discrete probability distribution
- Initially, Poisson distribution (mean=variance) was used, but this did not account for the observed overdispersion between biological replicates (counts from biological replicates have higher variance than the mean)
- RNAseq data can be better modelled using the **negative binomial distribution (NB)**; R packages DESeq2 and edgeR use NB distribution models
- Information of data distribution is important for statistical inference from a limited number of experimental replicates



Further reading: DESeq and DESeq2 papers
<https://doi.org/10.1186/gb-2010-11-10-r106>
<https://doi.org/10.1186/s13059-014-0550-8>

Differential gene expression analysis – details of the DESeq2 model

- Assumption: most genes are not differentially expressed
- Size factors s_j are incorporated into the negative binomial model to allow normalization by library size and RNA composition
- Log2-fold changes (LFC), e.g. change of genes between untreated and treated condition, are obtained from a generalized linear model

Count matrix K with $K_{i,j}$ sequencing reads

gene	sample_1	sample_2	sample...	sample_j
gene_a	count	count	count	count
gene_b	count	count	count	count
gene_c	count	count	count	count
gene_...	count	count	count	count
gene_i	count	count	count	count

Counts K_{ij} for gene i , sample j are modelled using a negative binomial distribution (NB)

$$K_{ij} \sim \text{NB}(\mu_{ij}, \sigma_{i,j}^2)$$

Mean scaled by **size factor** (normalization)

$$\mu_{ij} = s_j q_{ij}$$

LFC calculation using generalized linear model

$$\log_2(q_{ij}) = x_j \cdot \beta_i$$

μ_{ij} : mean

$\sigma_{i,j}^2$: variance

s_j : sample-specific size factor

q_{ij} : quantity proportional to the concentration of cDNA fragments from the gene in the sample

β_i : log2 fold change for gene i for each column of the model matrix X (e.g. treated or untreated sample)

DESeq2 normalization – DESeqCounts (with size factors)

1. Create a pseudo-reference sample (row-wise geometric mean)

gene	sampleA	sampleB	pseudo-reference sample
EF2A	1489	906	$\text{sqrt}(1489 * 906) = \mathbf{1161.5}$
ABCD1	22	13	$\text{sqrt}(22 * 13) = \mathbf{17.7}$

2. Calculate ratio of each sample to the pseudo-reference

gene	sampleA	sampleB	pseudo-reference sample	ratio of sampleA/ref	ratio of sampleB/ref
EF2A	1489	906	1161.5	$1489/1161.5 = \mathbf{1.28}$	$906/1161.5 = \mathbf{0.78}$
ABCD1	22	13	16.9	$22/16.9 = \mathbf{1.30}$	$13/16.9 = \mathbf{0.77}$

3. Calculate the normalization factor for each sample (size factor)

```
normalization_factor_sampleA <- median(c(1.28, 1.30))
```

```
normalization_factor_sampleB <- median(c(0.78, 0.77))
```

4. Calculate the normalized count values using the size factor

sampleA **median ratio** ~ 1.3
sampleB **median ratio** ~ 0.77

Raw Counts

gene	sampleA	sampleB
EF2A	1489	906
ABCD1	22	13

Normalized Counts

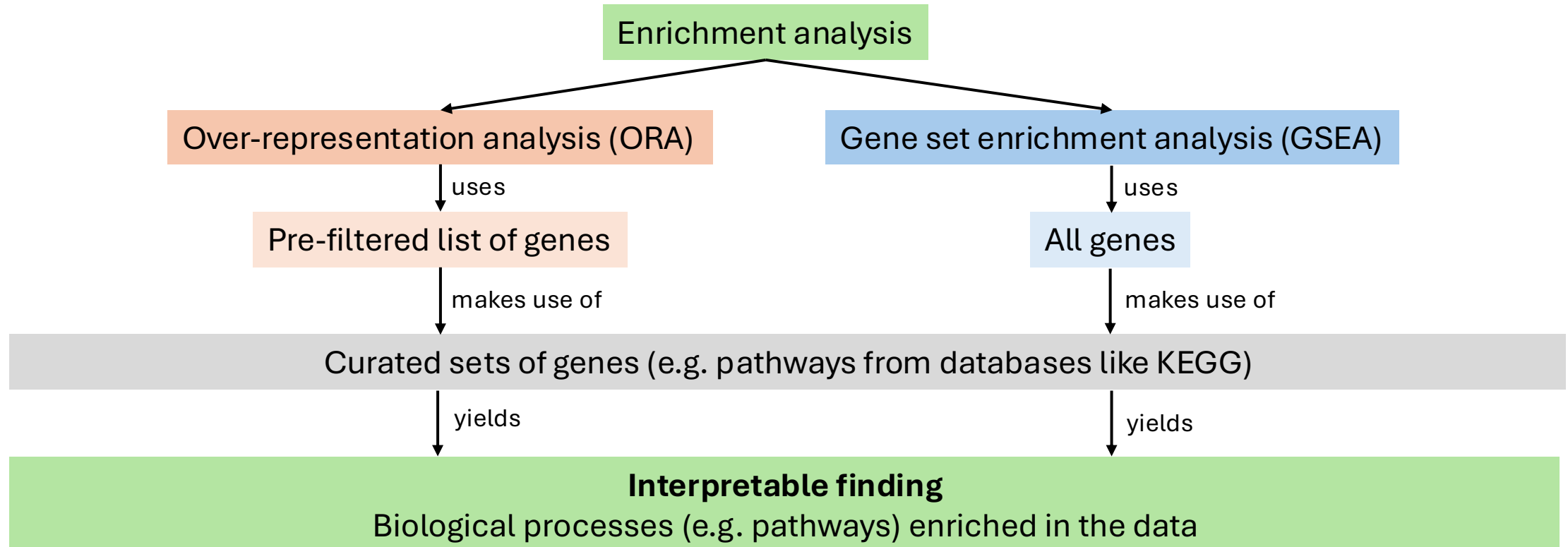
gene	sampleA	sampleB
EF2A	$1489 / 1.3 = \mathbf{1145.39}$	$906 / 0.77 = \mathbf{1176.62}$
ABCD1	$22 / 1.3 = \mathbf{16.92}$	$13 / 0.77 = \mathbf{16.88}$

Usually not performed “manually” as DESeq2 calculates it internally but we have to tell DESeq2 to do it

Functional analysis for biological interpretation

- **Enrichment analysis**

- Knowledge-based approach for biological interpretation of omics data, e.g. RNAseq results
- **Goal:** find biological terms (curated gene sets for biological processes) that are enriched in the data for biological interpretation (no single-gene interpretation)
- Biological term/pathway lists are obtained from KEGG, GO etc or can be manually curated (often .gmt files)
- Different types of enrichment analysis



Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA)** – all genes ranked by fold-change and p-value

List of all genes

	Gene symbol	p-val	Differentially expressed	FC
1	<i>HAO1</i>	0.0001	Yes	-5.6
2	<i>RDH16</i>	0.00045	Yes	-7.6
3	<i>ALDOB</i>	0.00009	Yes	-6.7
4	<i>AKR1C4</i>	0.0051	Yes	-10.0
5	<i>ABCB11</i>	0.067	Yes	10.0
6	<i>BAAT</i>	0.0034	Yes	23.5
7	<i>SLC2A2</i>	0.00990	Yes	2
8	<i>FCER1G</i>	0.00673	Yes	6.7
...				
6532	<i>OSM</i>	0.00009	Yes	9
6533	<i>VEGFA</i>	0.067	Yes	8.5
6534	<i>ZAP70</i>	0.0034	Yes	7.8

$$\text{RANKING} = \text{SIGN}(\text{FC}) * -\log_{10}(\text{p-value})$$

Ranked list of genes

	Gene symbol	Ranking
1	<i>JAK9</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>ALDOB</i>	11.3
5	<i>AKR1C4</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>BAAT</i>	3.6
8	<i>SLC2A2</i>	0.0099
9	<i>FCER1G</i>	0.00673
	...	
6532	<i>MyD88</i>	-9
6533	<i>TRIF</i>	-9.37
6534	<i>IRAK1</i>	-15.6

SIGNIFICANT + UPREGULATED

NON-SIGNIFICANT

SIGNIFICANT + DOWNREGULATED

Functional analysis for biological interpretation

- **Gene-set enrichment analysis (GSEA)** – Which biological processes are enriched at the top and at the bottom of the ranked gene list?

	Gene symbol	Ranking
1	<i>PTPN22</i>	12.9
2	<i>RDH16</i>	11.4
3	<i>IL1B</i>	11.3
5	<i>TNF</i>	10.7
6	<i>ABCB11</i>	8.6
7	<i>SPON2</i>	3.6
8	<i>SIC2A2</i>	0.0099
9	<i>TRIM55</i>	0.00673
	...	
6532	<i>OSM</i>	-9
6533	<i>TLR6</i>	-9.37
6534	<i>ZAP70</i>	-15.6

RANKED LIST



(181 GENES)

interleukin-6 production
(GO:0032635)*

PTPN22, IL1B, CSK, TNF,
TRAF6, TLR2, RABGEF1,
PYCARD, SIRPA, SPON2,
TLR7, TLR6, TRIM55, ZG16,
TYROBP, UCN, XBP1...

GENE SET

*GO = Gene Ontology

Result will show genes and biological terms that are

Up-regulated

[illegible]

Down-regulated

[illegible]

Not enriched

[illegible]

Statistical metrics:

- Enrichment score (ES) of a biological term represents the running sum when walking down the list
- Normalised enrichment score (NES): ES is normalised to make biological terms with different number of genes comparable
- p-value from permutations (adjusted)

Functional analysis for biological interpretation

- **Enrichment analysis – details and summary**

	Over-representation analysis (ORA)	Gene set enrichment analysis (GSEA)
Input gene list	<u>Filtered list</u> , e.g. $p_{adj} < 0.05$ and $\log_2FC > 1 $ → Genes not passing the thresholds will be missed	<u>Entire gene list ranked</u> by fold-change (with sign) and/or p_{adj} → No pre-filtering of genes is an advantage
Separation of up- and down-regulated genes	Can be done	No
Core statistical result and tests used	• Typically an adjusted p value • Fisher's exact test (one-sided), hypergeometric distribution	• Normalized enrichment score (NES) and adjusted p value • Modified Kolmogorov-Smirnov and permutation test
Software packages (selection)	mulea, g:profiler, Enrichr, clusterProfiler, GSEApy (via Enrichr)	fgsea, clusterProfiler, GSEApy

RNAseq data analysis hands-on

- R scripts in codspace (Juliana)

Wrap up and final Q&A

Thanks for attending!

Feel free to contact us for help in RNAseq or other omics processing and analyses projects!

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Benefiting people and society